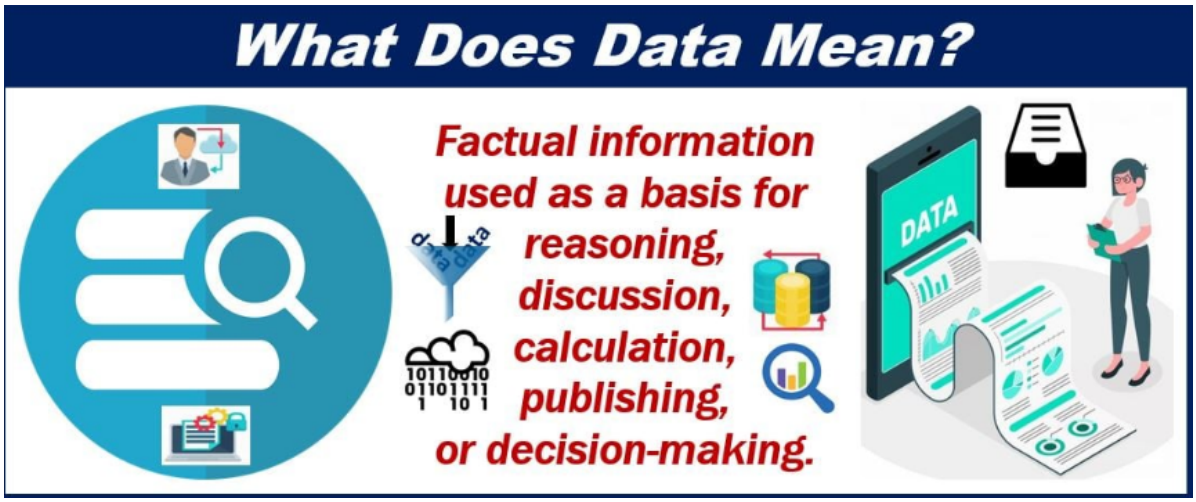
CONCEPT

Definition of Data

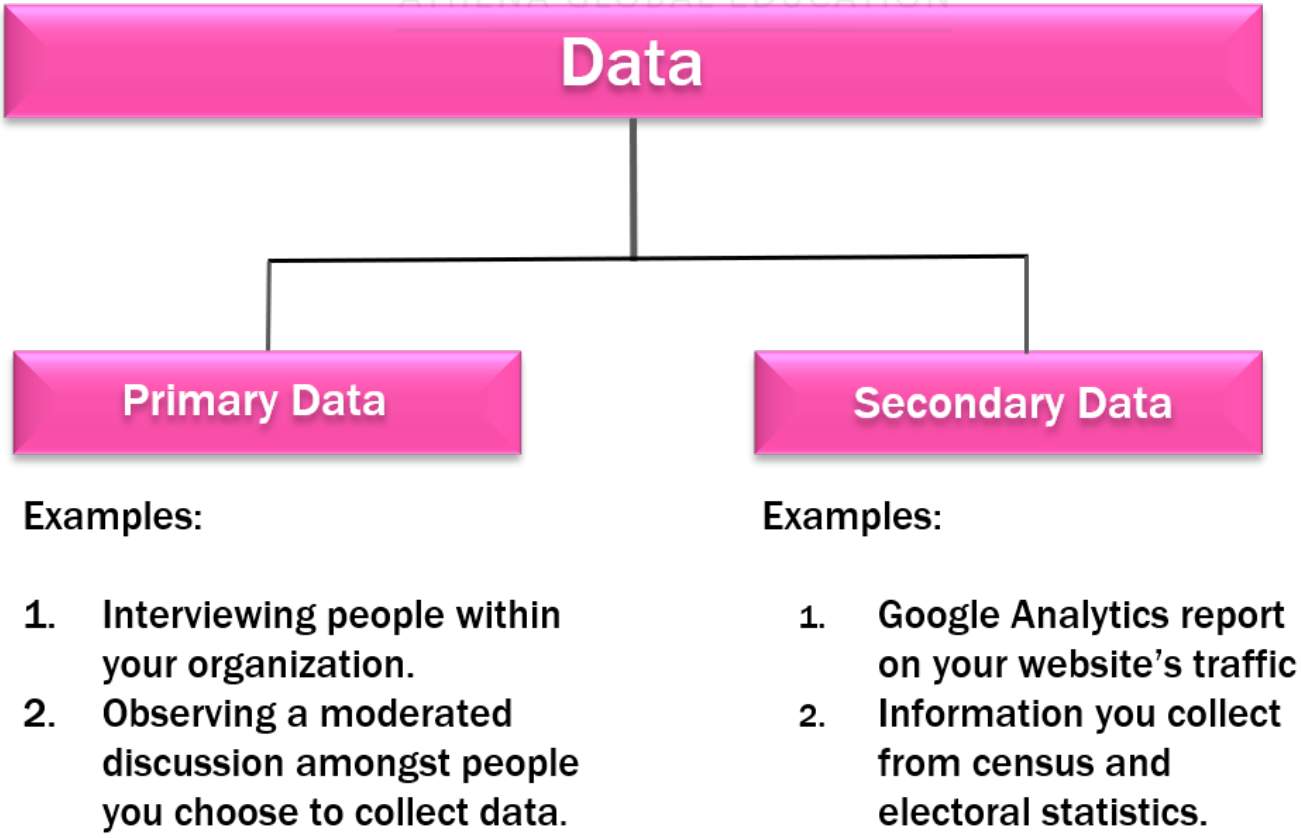
Data are facts that can be distinguished. They may be composed of well-formed models or terms.

Data are factual information (such as measurements or statistics) used as a basis for reasoning, discussion, or calculation



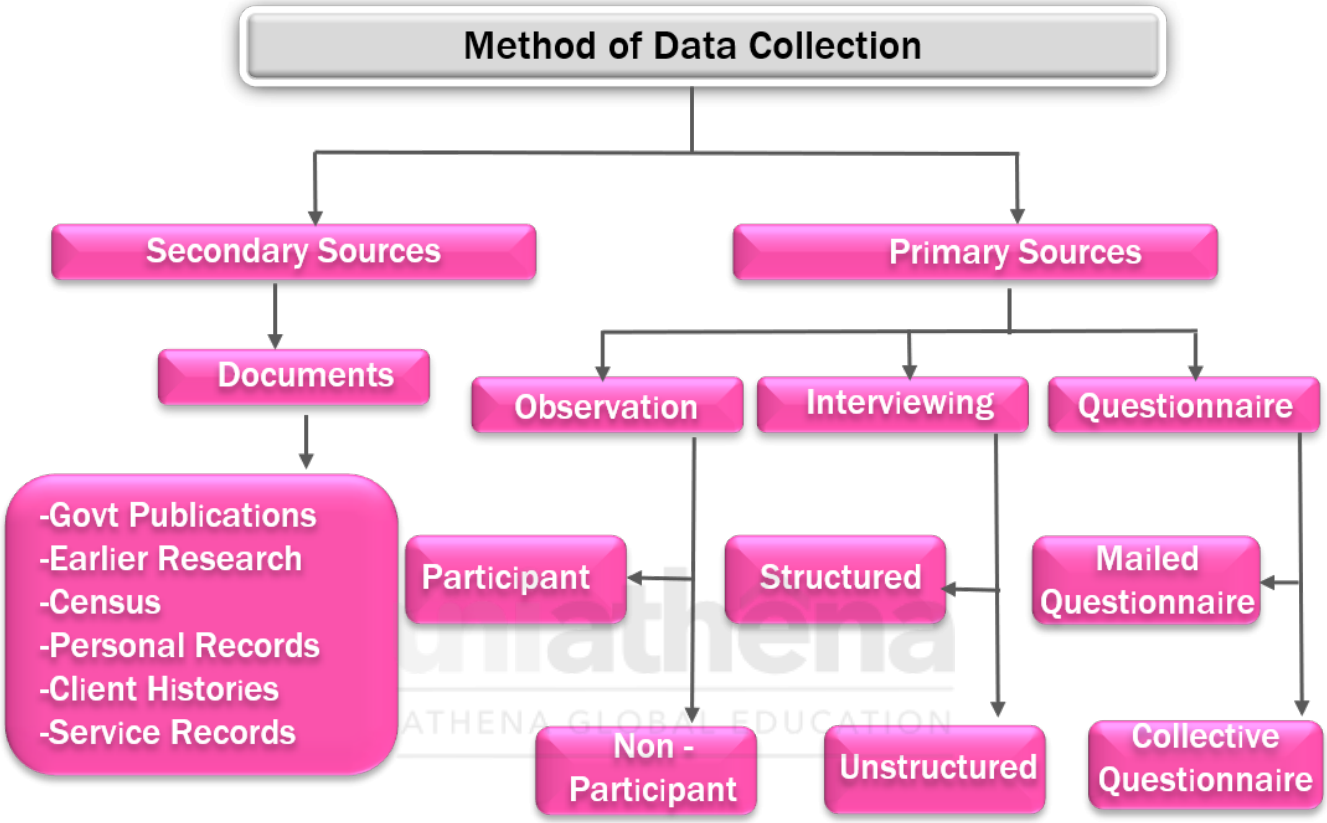
Source: marketbusinessnews.com

Types of Data



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- **Primary data** is what we usually mean by data: the content of documents or databases, or messages transferred over a network, or sensor readings.
- **Metadata** refers to primary data by describing its format, version, status, and so forth. Metadata is data about data.
- **Operative Data** is a subtype of metadata describing how data is to be used.
- **Derivative Data** is derived from the three above-mentioned data types. It can be used to sort, organize, or search for patterns in the above-mentioned types of data.



Definition of Information

Information is **well-formed data with meaning**. Data must follow syntactical rules. Information goes beyond that by adding meaning. There is a reference to the **"well-formedness"** of the combined data elements because meaning cannot be assigned to terms that violate syntax rules.

Different data can mean the same thing. Vice versa, the same data can mean different things - if it is interpreted differently. Because information consists of data and assigned meaning, different data can never be the same information (but can mean the same thing).

Knowledge

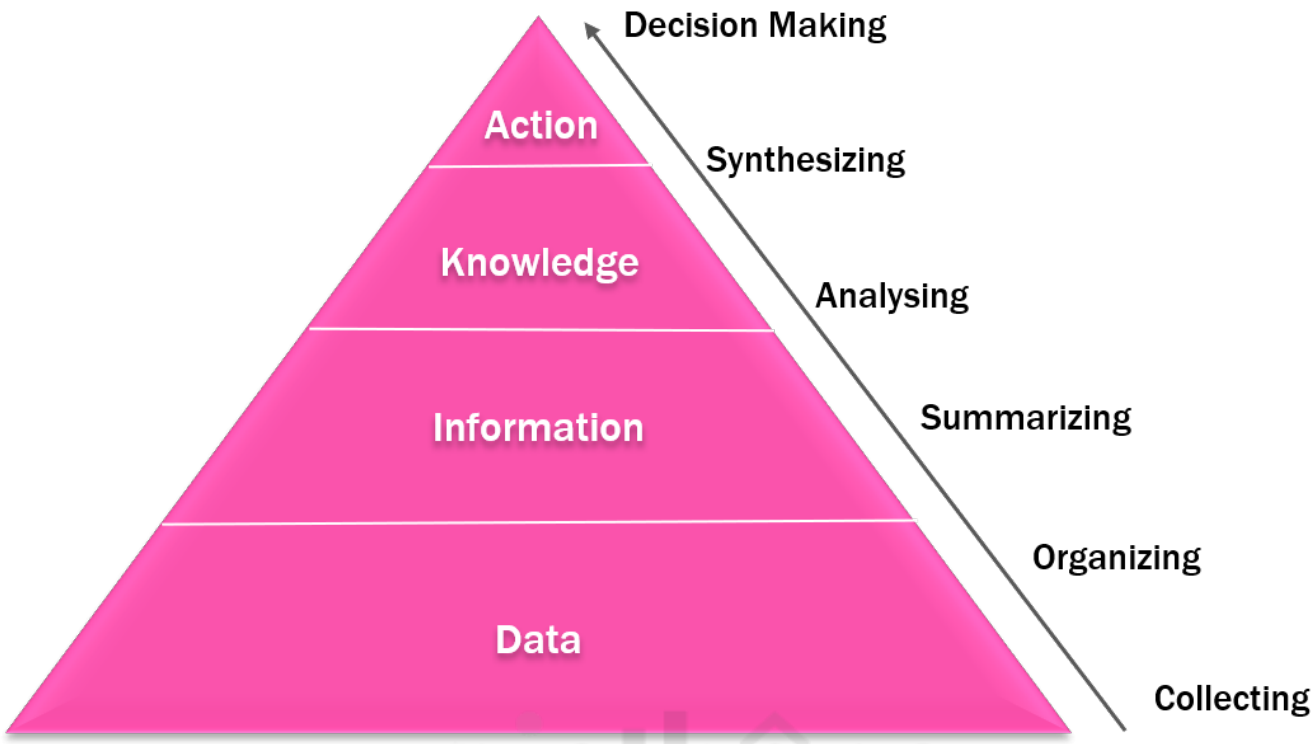
"Knowledge is the human expertise stored in a person's mind, gained through experience and interaction with the person's environment".

This definition provides a link to interaction and experience, and it emphasizes: knowledge resides in the mind. When knowledge management refers to the organizational dimension of learning, knowledge must be spread. In the sense of this can only mean converting the knowledge of person A into information and finally into data. Data can be stored and information re-created by again

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assigning meaning and context.

Data, Information, Knowledge, Action



Information and knowledge are the outcomes of human action that engage signs, signals, and artifacts in social and physical settings. **Knowledge** builds on an accumulation of experience. **Information** depends on an aggregation of data. The transformation of data into information and then into knowledge requires a human cognitive effort that results in the perception of structure and the attribution of meaning and significance. Consider a document containing a table of numbers indicating product sales for the quarter. As they stand, these numbers are data. An executive reads these numbers, recognizes the name and nature of the product, and notices that the numbers are lower than the corresponding figures of the last three quarters, thus indicating a downward trend. **Data** has become information. The individual considers possible explanations for the product decline and concludes that the product is no longer attractive to its customer. This new belief, derived from reasoning and reflection, is knowledge. **Based on this knowledge, the individual may decide to take some action** (such as adding new product features or identifying new markets). In this way, information is data given context and vested with meaning and significance.

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